

Evolution: Most Expected Board Questions & Answers

Comprehensive Standard Blueprint • 1, 2, 3 & 5 Mark Solved Questions • Master Review

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PART A: VERY SHORT ANSWER QUESTIONS (1 MARK EACH)

1 Mark Q1. Define organic evolution.

Ans: Organic evolution is the gradual, continuous, and progressive development of complex organisms from simpler ancestral life forms over geological epochs ("descent with modification").

1 Mark Q2. What is biopoiesis?

Ans: Biopoiesis is the natural process by which living cells or organisms first originated from inanimate, non-living matter on early Earth (chemical evolution).

1 Mark Q3. Who proposed the Big Bang Theory? Mention the age of universe.

Ans: Proposed by **Abbé Georges Lemaître** (1927). The universe is estimated to be approximately **20 billion years old**.

1 Mark Q4. Name the gases used by Stanley Miller in his spark discharge simulation.

Ans: Methane (CH_4), Ammonia (NH_3), and Hydrogen (H_2) in the ratio of **2 : 1 : 2** along with water vapor (H_2O) at **800°C**.

1 Mark Q5. State the Biogenetic Law proposed by Ernst Haeckel.

Ans: "*Ontogeny recapitulates phylogeny*" — the embryonic developmental stages of an individual organism repeat the evolutionary adult ancestral history of its race.

1 Mark Q6. Who disproved Haeckel's Biogenetic Law?

Ans: **Karl Ernst von Baer** disproved it by proving that embryos never pass through the adult stages of ancestral animals, but only share generalized embryonic archetypes.

1 Mark Q7. What is saltation as described by Hugo de Vries?

Ans: Saltation refers to a **single-step large mutation** that brings about rapid, discontinuous speciation.

1 Mark Q8. Name the connecting link between reptiles and birds.

Ans: *Archaeopteryx lithographica*.

1 Mark Q9. What are coacervates? Who proposed this concept?

Ans: Coacervates are micro-colloidal droplets formed by the aggregation of protein-like substances surrounded by a water membrane shell, proposed by **A.I. Oparin (1938)**.

1 Mark Q10. What is meant by the Founder Effect?

Ans: When a small splinter group of a population drifts away to establish a new isolated colony, the resulting drastic shift in allele frequency makes them distinct from the original gene pool.

1 Mark Q11. Name the living fossil fish caught in South Africa in 1938.

Ans: **Coelacanth** (*Latimeria chalumnae*), belonging to the lobe-fish (Crossopterygii).

1 Mark Q12. State the cranial capacity of Neanderthal man.

Ans: Approximately **1400 cc** (lived 100,000 to 40,000 years ago).

PART B: SHORT ANSWER QUESTIONS (2 MARKS EACH)

2 Marks Q13. Differentiate between Homologous and Analogous organs.

Homologous Organs	Analogous Organs
Same origin and anatomical build; different functions.	Different origin and anatomy; perform similar functions.
Lead to Divergent Evolution .	Lead to Convergent Evolution .
Ex: Forelimbs of whale and bat.	Ex: Wings of butterfly and bird.

2 Marks Q14. How did Louis Pasteur disprove the Theory of Spontaneous Generation?

Ans: Pasteur used custom **Swan-necked (S-shaped) flasks** containing nutrient broth. The curved neck permitted free passage of air while trapping dust and airborne microbes in the bend. The boiled broth remained sterile indefinitely until the neck was broken or tilted, proving that living organisms arise exclusively from pre-existing life (Biogenesis).

2 Marks Q15. Write any two reptilian and two avian characters of *Archaeopteryx*.

Ans:

- **Reptilian characters:** Presence of teeth in jaws, clawed free fingers on wings, elongated tail with unfused caudal vertebrae.
- **Avian characters:** Forelimbs modified into feathered airfoil wings, jaws modified into a beak, presence of contour feathers.

2 Marks Q16. What is adaptive radiation? Give two classic examples.

Ans: Adaptive radiation is the evolutionary diversification of an ancestral species radiating from a single geographical point into different specialized ecological niches.

- *Example 1:* Darwin's finches in the Galapagos Islands.
- *Example 2:* Australian marsupials evolving into various forms on the isolated Australian continent.

2 Marks Q17. Explain August Weismann's experiment to disprove Lamarckism.

Ans: Weismann conducted the **decaudalization experiment** by surgically amputating the tails of mice for 22 consecutive generations. In every generation, only normal-tailed progeny were born. This disproved the inheritance of acquired somatic characters and led to the formulation of the **Germplasm Theory** (only germinal variations are heritable).

2 Marks Q18. Distinguish between Darwinian variation and Mutation theory.

Darwinian Variation	De Vries' Mutation
Small, continuous, and directional.	Large, sudden, random, and directionless.
Causes gradual speciation over epochs.	Causes single-step speciation (saltation).

PART C: ANALYTICAL QUESTIONS (3 MARKS EACH)

3 Marks Q19. Describe Industrial Melanism in *Biston betularia* as an example of natural selection.

Ans:

- Before Industrialization (1850s):** Tree trunks were clad in light-colored lichens. The white-winged moths (*typica*) camouflaged against tree bark and survived predation, whereas the dark melanic mutants (*carbonaria*) were heavily predated.
- After Industrialization (1920s):** Coal soot blackened tree trunks and killed sensitive lichens. Consequently, dark melanic moths blended invisibly and survived, while white moths were easily detected and preyed upon.
- Inference:** Nature favored the phenotype endowed with adaptive camouflage; neither variant was completely eliminated.

3 Marks Q20. State the Hardy-Weinberg Principle and explain any four factors that upset it.

Ans: Principle: In a large, randomly mating population, allele frequencies remain constant from generation to generation in the absence of evolutionary disturbances ($p^2 + 2pq + q^2 = 1$).

Disturbing Factors:

- **Gene Flow:** Immigration or emigration altering allele frequencies.
- **Genetic Drift:** Random, chance fluctuations in small gene pools (Founder effect and Bottleneck effect).
- **Mutations:** Introducing new alleles into the gene pool.
- **Recombination:** Crossing over during meiosis generating new allele combinations.

3 Marks Q21. Explain the three operational types of Natural Selection with curve shifts.

Type	Selection Action	Phenotypic Curve Shift
1. Stabilizing	Favors mean intermediate phenotype; eliminates extremes.	Peak becomes higher and narrower (Ex: Human infant birth weight).
2. Directional	Favors one extreme phenotype over others.	Peak shifts in one direction (Ex: Industrial melanism, antibiotic resistance).
3. Disruptive	Favors both extreme phenotypes; eliminates mean.	Splits curve into two distinct peaks (bimodal) (Ex: Rabbit fur on black/white rocks).

3 Marks Q22. Compare the convergent radiation between Australian Marsupials and Placental Mammals.

Ans: Isolated in similar ecological niches through continental drift, distinct lineages evolved parallel adaptive specializations:

Placental Mammal	Australian Marsupial Counterpart
Anteater	Numbat (banded anteater)
Flying squirrel	Sugar glider (flying phalanger)
Bobcat	Tasmanian tiger cat
Wolf	Tasmanian wolf (Thylacine)

PART D: LONG ESSAY QUESTIONS (5 MARKS EACH)

5 Marks Q23. Describe Stanley Miller's spark discharge experiment on chemical origin of life with a neat labeled diagrammatic schematic.

Experimental Design & Methodology:

- **Simulation of Prebiotic Atmosphere:** Stanley Miller and Harold Urey constructed a closed glass Spark Discharge Apparatus to recreate conditions prevailing on prebiotic Earth.
- **Gaseous Mixture:** Introduced a reducing gas mixture of Methane (CH_4), Ammonia (NH_3), and Hydrogen (H_2) in the volumetric ratio of **2 : 1 : 2**.

Observation, Analysis & Inferences:

- **Observation:** Paper chromatography revealed the abiogenic synthesis of complex organic compounds, predominantly racemic **amino acids (glycine, alanine, aspartic acid)**, urea, fatty acids, hydrogen cyanide, and organic acids.
- **Significance / Scientific Inference:** Proved experimentally that organic biomolecules—the building blocks of living systems—can be formed spontaneously

- **Simulated Ocean:** Heated water in a round-bottom flask generating steam that circulated through the gas mixture.
- **Energy Source:** Two tungsten electrodes provided continuous electric sparks (~75,000 V) at **800°C** to simulate lightning discharge.
- **Condensation:** The circulated vapor passed through a water condenser and condensed into a liquid trap for 7 days.

from simple inorganic gases under reducing prebiotic environmental conditions (validating the **Oparin-Haldane Hypothesis**).

Board Exam Diagram Checkpoints: Ensure you label the: (1) Electrodes, (2) Spark discharge chamber, (3) Boiling water flask, (4) Condenser (water-in/water-out), and (5) Trap containing synthesized organic molecules.

5 Marks Q24. Give a comprehensive analytical account of the Origin and Evolution of Man with evolutionary chronology, cranial capacities, and distinctive cultural characteristics.

Hominid Stage	Chronological Era	Cranial Capacity	Key Anatomical, Dietary & Cultural Milestones
1. <i>Dryopithecus africanus</i> (<i>Proconsul</i>)	~15 – 20 mya (Miocene)	~350 – 400 cc	Arboreal, ape-like knuckle walker; arms and legs equal length; frugivorous; common ancestor of man and great apes.
2. <i>Ramapithecus punjabicus</i>	~14 – 15 mya (Late Miocene)	~400 – 450 cc	Fossils discovered in Siwalik Hills (India) by Edward Lewis; parabolic dental arcade; upright bipedal stride; more man-like than <i>Dryopithecus</i> .
3. <i>Australopithecus africanus</i> ("Taung Baby", "Lucy")	~2 – 4 mya (Pliocene)	450 – 600 cc	First obligate bipedal posture; lumbar lordosis; lived in East African grasslands; hunted with stone weapons; fruit-eaters.
4. <i>Homo habilis</i> ("Handy Man" / Tool maker)	~1.6 – 2.0 mya (Pleistocene)	650 – 800 cc	First human-like hominid; crafted Oldowan stone pebble tools; omnivorous (probably did not eat large mammal meat).
5. <i>Homo erectus</i> (Java Man, Peking Man)	~1.5 mya (Mid-Pleistocene)	~900 cc	Fossils in Java (Eugene Dubois); fully erect posture; crafted Acheulean handaxes; controlled fire; regular meat-eaters.
6. <i>Homo neanderthalensis</i> (Neanderthal Man)	100,000 – 40,000 YA	~1400 cc	Heavy brow ridges, stooped robust build; built cave shelters; wrapped bodies in animal hides; practiced ceremonial burials with flowers.
7. <i>Homo sapiens fossilis</i> (Cro-Magnon Man)	34,000 – 10,000 YA	~1650 cc	Direct ancestor of modern man; tall (~1.8 m), prominent orthognathic chin; Aurignacian bone tools; painted prehistoric cave art (Lascaux).
8. <i>Homo sapiens sapiens</i> (Modern Human)	75,000 YA – Present	1350 – 1450 cc	Arose in Africa during ice ages; global radiation; developed modern language, agriculture, and human settlements ~10,000 years ago.

5 Marks Q25. Write an essay on Darwin's Theory of Natural Selection. Highlight the salient postulates, Wallace's contribution, and its major scientific criticisms.

Salient Postulates of Darwinism:

- **Overproduction (Enormous Fecundity):** All living organisms possess a geometric reproductive capacity to multiply exponentially.
- **Limitation of Food and Space:** Environmental resources remain relatively constant, creating an inevitable **struggle for existence** (intra-specific, inter-specific, and environmental).

Wallace's Contribution & Criticisms:

- **Alfred Russel Wallace:** Working independently in the Malay Archipelago, arrived at identical conclusions regarding natural selection and co-authored the 1858 joint paper.
- **Key Criticisms of Darwin's Theory:**
 1. Failed to explain the source and origin of heritable genetic variations (lack of modern genetic knowledge).

- **Variations:** No two individuals in a population are identical. Useful variations endow individuals with greater environmental fitness.
 - **Survival of the Fittest (Natural Selection):** Herbert Spencer coined "survival of the fittest," which Darwin formulated as Natural Selection — the differential reproductive survival of variants endowed with adaptive advantages.
 - **Branching Descent & Speciation:** Accumulation of favorable variations over long periods leads to the origin of new species.
2. Could not explain the survival vs. **arrival of the fittest**.
 3. Did not account for over-specialized organs (e.g., massive Irish elk antlers, elephant tusks).
 4. Could not explain the persistence of vestigial, non-functional organs in animals.